

19. Which of the following is used as an immuno suppressor for kidney transplantation?

- (a) Antibiotics (b) Vaccines
(c) Interferon (d) Monoclonal antibodies

Uttarakhand P.C.S. (Pre) 2024

Ans. (d)

Immunosuppressants (anti-rejection medicines), are medicines that keep kidney transplants from being attacked by the immune system. Monoclonal antibodies play a crucial role in kidney transplant immunosuppression. A series of monoclonal antibodies (mAbs) are commonly utilized in kidney transplantation as induction therapy (a period of intense immunosuppression immediately before and following the implant of the allograft), to treat steroid-resistant acute rejections, to decrease the incidence and mitigate effects of delayed graft function, and to allow immunosuppressive minimization. They target specific cells or molecules involved in the immune response, like T cells or the IL-2 receptor.

VII. Nervous System

Notes

- The **nervous system** is a complex collection of nerves and specialized cells known as neurons that transmit signals between different parts of the body. It is essentially the body's **electrical wiring**.
- In other words, we can say that a nervous system is a group of organized cells specialized in the conduction of electrochemical stimuli from sensory receptors through a network to the site at which a response occurs.
- A neuron or nerve cell is a cell that carries electrical impulses.
- Neurons** are the basic unit of the nervous system.
- Every neuron is made of a cell body (also called a soma), dendrites and a single axon.
- Dendrites and axons are nerve fibres.
- Nerve cells are the longest cell of the body.
- A neuron is an **electrically excitable cell** that communicates with another cell via specialized connections called **synapses**.
- All animals except sponges and placozoans have neurons, but other multicellular organisms such as plants do not.
- Neurons fall into following types –
 - Sensory neurons** (Afferent neurons) respond to stimuli such as touch, sound or light that affect the cells of sensory organs and sends signals to the spinal cord or brain.

- Motor neurons** (Efferent neurons) receive signals from the brain and spinal cord and carries it toward the peripheral effector organs (mainly muscles and glands).

- Interneurons** connect neurons to other neurons within the specific regions of the brain or spinal cord in neural networks.

- In vertebrates, the nervous system has two major parts –

(i) Central Nervous System (CNS)

(ii) Peripheral Nervous System (PNS)

- The Central Nervous system (CNS) consists of the **brain** and **spinal cord**.
- The Peripheral Nervous system (PNS) consists mainly of nerves, which are enclosed bundles of the long fibres or axons, that connect the CNS to every other part of the body.
- The peripheral nervous system is further subdivided into the **somatic**, **autonomic** and **enteric** nervous system.

(a) Somatic Nervous System :

- The somatic Nervous system is also called **voluntary nervous** system.
- It is the part of the peripheral nervous system associated with the voluntary control of body movements via skeletal muscles. The somatic nervous system consists of afferent nerves or sensory nerves, and efferent nerves or motor nerves.

(b) Autonomic nervous system :

- The part of the peripheral nervous system that controls and regulates the internal organs without any conscious recognition or effort by the organism.
- The autonomic nervous system comprises two antagonistic sets of nerves.

(i) Sympathetic – Connects the internal organs to the brain by spinal nerves, when stimulated, these nerves prepare the organisms for stress by increasing the heart rate, increasing blood flow to the muscles and decreasing blood flow to the skin. The sympathetic nervous system is activated in case of emergencies to mobilize energy.

(ii) Parasympathetic – The nerve fibres of the parasympathetic nervous system are the **cranial nerves**, primarily the **vagus nerve** and the lumbar **spinal** nerve. When stimulated these nerves increase digestive secretions and reduce the heartbeat. The parasympathetic nervous system is activated when organisms are in relaxed state.

- Within the brain, the autonomic nervous system is regulated by the **hypothalamus**. The autonomic nervous system controls internal body processes such as following-
 - Blood pressure
 - Heart and breathing rates

- Body temperature
- Digestion
- Body metabolism (thus affecting body weight)
- The balance of water and electrolytes (such as sodium and calcium)
- The production of body fluids (saliva, sweat & tears)
- Urination
- Defecation
- Sexual response
- Many organs are controlled primarily by either the sympathetic or the parasympathetic division. Sometimes the two divisions have opposite effects on the same organs. For example - the sympathetic division increases blood pressure and the parasympathetic division decreases it. Overall, the two divisions work together to ensure that the body responds appropriately to different situations.
- The two chemical messengers (neurotransmitters) are used to communicate within the autonomic nervous system :
 - Acetylcholine
 - Norepinephrine (Noradrenaline)
- Nerve fibres that secrete acetylcholine are called **cholinergic fibres** and which secrete norepinephrine are called **adrenergic fibres**.
- Generally, acetylcholine has parasympathetic (inhibiting) effects and norepinephrine has sympathetic (stimulating) effects. However, acetylcholine has some sympathetic effects too. For example - it sometimes stimulates sweating or makes the hair stand.

(c) Enteric nervous system :

- It consists of a mesh-like system of neurons that governs the function of the gastrointestinal tract. It is capable of acting independently of the sympathetic and parasympathetic nervous system, although it may be influenced by them. It is also called the **second brain**.
- Some scientists include the enteric nervous system as part of autonomic nervous system.

Cranial and Spinal nerves :

- Nerves that exit from the cranium are called **cranial nerves** while those exiting from the spinal cord are called **spinal nerves**.
- The **cranial nerves** are 12 pairs of nerves which are found on the ventral (bottom) surface of the brain.
- Cranial nerves relay information between the brain and parts of the body, primarily to and from regions of the head and neck.
- A **spinal nerve** is a mixed nerve, which carries motor, sensory, and autonomic signals between the spinal cord and the body. In the human body there are 31 pairs of spinal nerves, one on each side of the vertebral column.
- The cranial nerves are considered components of the peripheral nervous system although on a structural level

the olfactory (I), optic (II), and trigeminal (V) nerves are more accurately considered part of the central nervous system.

- The spinal nerves are parts of the peripheral nervous system.

Human Brain :

- The human brain is the command centre of the human nervous system.
- The brain is one of the largest and most complex organs in the human body.
- The brain consists of - the **cerebrum**, the **cerebellum** and the **brainstem** (which is composed of the midbrain, the pons, and the **Medulla oblongata**).
- The **cerebrum** is the largest part of the brain and is composed of the right and left hemisphere. It contains the cerebral cortex, as well as several subcortical structures, including the hippocampus, basal ganglia, and olfactory bulb. It performs higher functions like interpreting touch, smell, vision and hearing, as well as speech, reasoning, emotions, learning and fine control of movement.
- **Cerebellum** receives information from the sensory systems, the spinal cord and other parts of the brain and then regulate motor movement. It coordinates voluntary movements such as postures balance, coordination and speech, resulting in smooth and balanced muscular activity.
- **Medulla oblongata** is the lower part of the brainstem where the brain transitions to the spinal cord. It is only about 3 cm. long but the medulla is an indispensable nerve tract which contains the control centres for our autonomic vital functions – heart rate, blood pressure, breathing, vomiting, coughing, sneezing and many involuntary reflexes.

Reflex Actions :

- When you touch a hot object or when a pin picks your finger, what is your immediate reaction? Of course, you remove your hand away from the source of pain, either the hot object or the pin. In situations like these, your reactions are always immediate, involuntary and sudden. These happen without much thinking process. In scientific terms, this action is called the **reflex action**. Here, the **spinal cord** has a major role to play.
- The whole mechanism of reflex action occurs in such a fashion that there is no conscious control of brain. Stimulation occurs through the peripheral nervous system and the response to this peripheral nerve stimulation is involuntary. In a reflex action, the spinal cord along with brainstem is responsible for the reflex movements.
- The two extremes in the animal kingdom are **endothermic** homeotherms and **ectothermic** poikilotherms.

- Most **mammals**, including humans as well as most birds are endothermic homeotherms – maintains a stable internal body temperature regardless of external influence, while most fish, invertebrates, reptiles and amphibians are ectothermic poikilotherms—whose internal temperature varies considerably.
- In humans, the average normal internal temperature is 98.6°F (37°C or 310K).

Question Bank

1. **There are how many pairs of cranial nerves in human?**

- (a) 8 (b) 12
(c) 25 (d) 31

M.P. P.C.S. (Pre) 2020

Ans. (b)

Nerves that exit from the cranium are called cranial nerves. There are 12 pairs of cranial nerves which are found on the ventral (bottom) surface of the brain.

2. **How many pairs of nerves arise from the spinal cord?**

- (a) 13 (b) 31
(c) 33 (d) 12

45th B.P.S.C. (Pre) 2001

Ans. (b)

Nerves that arise from the spinal cord are called spinal nerves. A spinal nerve is a mixed nerve, which carries motor, sensory and autonomic signals between the spinal cord and the body. In human there are 31 pairs of spinal nerves, one on each side of the vertebral column. These are grouped into the corresponding cervical, thoracic, lumbar, sacral and coccygeal regions of the spine. There are eight pairs of cervical nerves, twelve pairs of thoracic nerves, five pairs of lumbar nerves, five pairs of sacral nerves and one pair of coccygeal nerves. The spinal nerves are part of the peripheral nervous system.

3. **Input and output nerves meet at :**

- (a) liver (b) central nervous system
(c) heart (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

The central nervous system has four parts, namely, Brain, Spinal cord, Nerve and Neurons. The Input and output nerves meet at several places, particularly in the spinal cord. Nerves from all over the body meet in a bundle in the spinal cord on their way to the brain. Reflex arcs are formed in this spinal cord itself, although the information input also goes on to reach the brain. They do not meet in the liver or the heart as they do not form part of nervous system.

4. **The brain is responsible for :**

- (a) Thinking (b) Regulating the heartbeat
(c) Balancing the body (d) All of the above three

U.P. P.C.S. (Pre) 2016

Ans. (d)

The brain is composed of the cerebrum, cerebellum, and brainstem. The cerebrum is the largest part of the brain and it performs higher functions like interpreting touch, vision and hearing as well as speech, reasoning, learning and fine control of movement. The function of the cerebellum is to coordinate muscle movements, maintain posture and balance. The brainstem performs many automatic functions such as breathing, heart rate, body temperature, wake and sleep cycles, digestion, etc. Hence, option (d) is the correct answer.

5. **Consider the following statements about human brain:**

1. **The brain is responsible for regulating the heart-beat.**

2. **It is responsible for balancing the body.**

Select the correct answer using the codes given below.

Code :

- (a) Only 1 is correct (b) Only 2 is correct
(c) 1 and 2 both are correct (d) Neither 1 nor 2 is correct

U.P. P.C.S. (Pre) 2020

Ans. (c)

See the explanation of above question.

6. **Where is the governing centre of reflex action located?**

- (a) In cerebrum (b) In cerebellum
(c) In spinal cord (d) In nerve cells

45th B.P.S.C. (Pre) 2001

Ans. (c)

A reflex arc is a neural pathway that controls an action reflex. In higher animals, most sensory neurons do not pass directly into the brain, but synapse in the spinal cord. This characteristic allows reflex actions to occur relatively quickly by activating spinal motor neurons without the delay of routing signals through the brain, although the brain will receive sensory input while the reflex action occurs. There are two types of reflex arc- autonomic reflex arc (affecting inner organs) and somatic reflex arc (affecting muscles).

7. **Which one of the following organs will not feel any pain on being pricked by a needle?**

- (a) Skin (b) Brain
(c) Heart (d) Eye

U.P.P.C.S. (Pre) 2017

Ans. (b)

There is no pain or tactile (touch) receptors of any type in the brain itself. So, a needle or even more invasive brain surgery is not associated with any 'sensation' of the brain. Brain does not have any pain receptors.

8. In a human brain memory power is found in :

- (a) Medulla oblongata (b) Cerebrum
- (c) Brain cavity (d) Cerebellum
- (e) All of these

Chhattisgarh P.C.S (Pre) 2013

Ans. (b)

The three main parts of the brain are the cerebrum, the cerebellum and the brainstem. The cerebrum is the biggest part of the brain and it is the part that controls movement, thinking, feeling and general problem-solving abilities. The cerebellum is smaller than the cerebrum. This part of the brain sits at the back of your skull, underneath the cerebrum. The cerebellum is responsible for controlling co-ordination and balance. The hippocampus is the part of the cerebrum that deals with memory.

9. 'Satiety' and 'thirst' centres in humans are located in which of the following parts of the brain ?

- (a) Fore-brain (b) Hypothalamus
- (c) Medulla (d) Optic lobe

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (b)

'Satiety' and 'thirst' centres in humans are located in the hypothalamus of the brain. This area of the brain governs body temperature, thirst, hunger, sleep, circadian rhythm, moods, sex drive and the release of various hormones in the body.

10. Body temperature is regulated by :

- (a) thalamus (b) hypothalamus
- (c) cerebellum (d) medulla
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

64th B.P.S.C. (Pre) 2018

Ans. (b)

The average normal temperature of the human body is 98.6° F. The hypothalamus controls body temperature. It also controls hunger, important aspects of parenting and attachment behaviours, thirst, fatigue, sleep and circadian rhythms.

11. The temperature of the human body :

- (a) Decreases in winters
- (b) Increases in summers
- (c) Neither decreases in winters nor increases in summers
- (d) Increases in winters

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (c)

The temperature of the human body neither decreases in winters nor increases in summers. The usual temperature of the human body is 98.6°F or 37°C and is controlled by the hypothalamus in the brain. The hypothalamus functions as a type of thermostat for the body.

12. The temperature of a normal human body is –

- (a) 98.4° F
- (b) 98° F
- (c) 98.8° F
- (d) None of the above

U.P.P.C.S. (Pre) 2007

Ans. (d)

The average of normal temperature of human body is 98.6° F (37° C or 310° K).

13. The normal temperature of the human body is –

- (a) 37° C (b) 98° C
- (c) 367 K (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

See the explanation of above question.

14. Which one of the following parts of the human brain is the regulating centre for swallowing and vomiting?

- (a) Cerebellum (b) Cerebrum
- (c) Medulla oblongata (d) Pons

I.A.S. (Pre) 2007

Ans. (c)

The medulla oblongata, which is only about three centimetre long, connects the higher levels of the brain to the spinal cord and is responsible for controlling several functions of the autonomous nervous system which include : reflex centre of vomiting, coughing, sneezing and swallowing. These reflexes which include the pharyngeal reflex, the swallowing reflex (also known as the palatal reflex) and the masseter reflex can be termed, bulbar reflexes.